Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/08/2026
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Project Title:	Reduced-Order Model and Experiments Using the PSI Data from STDCE		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2021 Physical Sciences NNH21ZDA014N-PSI: Use of the NASA Physical Sciences Informatics System – Appendix G
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Task Description:

It is proposed to utilize Physical Sciences Informatics (PSI) data of the Surface Tension Driven Convection Experiment (STDCE) to validate a low-dimensional model which takes interfacial deflections into account. This method will then be compared with a full numerical model that also tracks the interface deformation and flow dynamics of long-wave flows that accompany such experiments. The STDCE data is vital as it will be used via modern imagining software (eg., Track-Mate in Image J) to analyze flow profiles and thermal data. The full numerical model and the simplified low-order method will, upon validation, be employed to design and run a ground experiment whose objective is to use novel means via forced mechanical resonance to quench the long wave surface tension gradient flow.
The STDCE experiments were designed to understand the role of surface tension gradient-driven thermocapillary flow, often encountered in technologically relevant processes such as crystal growth. It was found that long wave steady two-dimensional convective flow gives way to oscillatory flow for certain parameter ranges. These long-wave flows are often detrimental to the relevant technological processes.

The advantage of using a reduced-order method in addition to the full numerical model is that interface evolution may be resolved via a semi-analytical and tractable set of equations for two state variables. This feature provides great insight into the physics of the convection and the root cause of the onset of oscillatory flow from a steady base flow. The nonlinearity of the model facilitates tracking of the interface deformation, which plays a significant role in the onset of oscillatory instability in STDCE. The PSI data will be invaluable as it will provide validation for the model by providing information from image analysis of that data.

The idea behind the proposed experiment is to generate thermal convection in a test-cell that emulates the STDCE and increase the forcing temperature difference until oscillatory flow is triggered. This test cell will be placed on a shaker, currently available in the Principal Investigator's laboratory, whose shaking frequency is set and amplitude is incrementally raised until the oscillatory flow is arrested and even the steady long wave flow is reduced. This is predicted to occur because resonant instabilities act on interfacial deformation and such deformation is characteristic of the STDCE. The critical amplitude to quench the undesired flow can be then compared with theory. The proposed ground-based experiments will identify operating parameter spaces where oscillatory instability may be suppressed. The semi-analytical model will also help identify the root causes for and suppression of oscillatory flow.

In summary, the PSI data is needed to validate the full numerical model and also the low-dimensional model which, in turn, allows us to extract the physical phenomena. The proposed ground experiment will provide additional validation for the model and also give information on suppression of convection via resonant forcing. The results of the study will be of great benefit to materials processing including crystal growth and in-space welding as well as additive manufacturing using directed energy deposition where free surface convection inevitably takes place.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research impacts are:
a) Low dimensional models that use length scale separation and which extract dominant physics in problems of fluid instability would be validated by complete models that do not make such approximations. This will allow low dimensional models to explain physical models with great simplifications. b) Obtaining a clear means on thwarting surface tension driven convection by means of resonant forcing.

The Earth Benefits are:

a) Potential technologies for improved heat transfer in micropipe heat exchangers b) Potential improvement for additive manufacturing

Task Progress:

We have two general aims in this project. The first is to validate low dimensional models at various levels of length scale ratios with Surface Tension Driven Convection (STDC) data, as the latter is well suited to such analysis. A second aim is to use low dimensional modeling to predict experimental behavior where we might strive to quench oscillatory Marangoni convection by enforcing suitable mechanical resonant forcing (i.e., by using Faraday forcing).
Progress Made in the First and Second Year

In the last three years we addressed these above aims and focused on the second aim to apply our knowledge of low-dimensional modeling. To this end, we studied a thin, laterally confined heated liquid layer subjected to mechanical parametric forcing without gravity. In the absence of parametric forcing, the liquid layer exhibits the Marangoni instability, provided the temperature difference across the layer exceeds a threshold. This threshold varies with the perturbation wavenumber, according to a curve with two minima, that correspond to long- and short-wave instability modes. The most unstable mode depends on the lateral confinement of the liquid layer. In wide containers, the long-wave mode is typically observed, and this can lead to the formation of dry spots. Our initial studies focus on this mode, as the short-wave mode is found to be unaffected by parametric forcing.

We used linear stability analysis and nonlinear computations, based on a reduced-order model, to investigate how parametric forcing can prevent the formation of dry spots. It was found that, at low forcing frequencies, the liquid film can be rendered linearly stable within a finite range of forcing amplitudes, which decreases with increasing frequency and ultimately disappears at a cutoff frequency. Outside this range, the flow becomes unstable to either the Marangoni instability (for small amplitudes) or the Faraday instability (for large amplitudes). At high frequencies, beyond the cutoff frequency, it was found that linear stabilization through parametric forcing is not possible. However, we showed that a nonlinear saturation mechanism, occurring at forcing amplitudes below the Faraday instability threshold, can greatly reduce the film surface deformation, and therefore prevent dry spots. Although dry spots can also be avoided at larger forcing amplitudes, this comes at the expense of generating large-amplitude Faraday waves.

In addition to understanding resonance in Newtonian fluids undergoing heating, we also studied the case of viscoelastic fluids. This is done because the silicone oil used in STDC has slight elastic properties. The results for the viscoelastic case stand in stark contrast with the Newtonian case. We now see an island of stability when resonance is applied to a viscoelastic fluid undergoing Marangoni convection. This island of stability becomes narrow as the elastic component increases, and this is attributed to the inertial effect that is afforded by memory.

Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Ignatius IB, Dinesh B, Dietze GF, Narayanan R. "Resonance-induced stabilization of Marangoni instability in viscoelastic fluids. " 12th Conference of the International Marangoni Association, Madrid, Spain, June 9-12, 2025. Abstracts. 12th Conference of the International Marangoni Association, Madrid, Spain, June 9-12, 2025. https://www.nebrija.com/nebrija-IMA12/abstracts-book-IMA12.pdf , Jun-2025
Articles in Peer-reviewed Journals	Ignatius IB, Dinesh B, Dietze GF, Narayanan R. "Thwarting Marangoni instability in viscoelastic liquid film via parametric forcing." Physical Review Fluids. 2025 Apr;10(4):044001. https://doi.org/10.1103/PhysRevFluids.10.044001 , Apr-2025
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